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HTML Course

Comprehensive Notes

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Topic 1: What is HTML

HTML stands for **HyperText Markup Language**. It is the standard language used to create and structure content on the web. Every website you visit is built with HTML at its core.

Key Concepts

- **HyperText** — Text that links to other text, enabling navigation between pages.
- **Markup** — Tags and elements used to annotate and structure content.
- **Language** — A defined set of rules and syntax for writing web content.
- HTML is **not** a programming language — it does not have logic or loops; it structures content.

HTML uses **elements** defined by tags. An element typically consists of an opening tag, content, and a closing tag:

```
<tagname> Content </tagname>
```

Tip: Understanding the Role of HTML

Think of HTML as the **skeleton** of a webpage. It provides structure, while CSS adds style and JavaScript adds behavior. Together, they form the foundation of every website.

Common Misconception

HTML is often confused with programming languages like Python or JavaScript. Remember: HTML **structures** content — it does **not** compute, calculate, or process logic.

Key takeaway: HTML is the foundation of all web pages. Learning it is the first step to becoming a web developer.

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Topic 2: HTML Structure and Elements

Every HTML document follows a standard structure that browsers rely on to render content correctly. Understanding this structure is essential for writing valid, well-formed HTML.

Key Concepts

- `<!DOCTYPE html>` — Declares the document type (HTML5).
- `<html>` — The root element wrapping all content.
- `<head>` — Contains metadata: title, charset, links to stylesheets.
- `<body>` — Contains all visible page content.
- `<meta charset="UTF-8">` — Ensures proper character encoding.
- `<title>` — Sets the browser tab title.

The Basic HTML Template:

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>My Page</title>
</head>
<body>
<!-- Content here -->
</body>
</html>
```

Tip: Always Start with the Template

Use this basic template as the starting point for **every** HTML file. It ensures your page is properly structured and rendered by all browsers.

Warning: Never Skip DOCTYPE

Without the `<!DOCTYPE html>` declaration, browsers may enter **quirks mode**, causing unpredictable rendering behavior.

Key takeaway: A well-structured HTML document always includes DOCTYPE, html, head, and body elements in the correct order.

Topic 3: Text and Formatting

HTML provides a rich set of elements for displaying and formatting text. These elements control how content appears semantically and visually in the browser.

Key Concepts

- `<h1>` to `<h6>` — Heading levels (h1 is largest/most important).
- `<p>` — Paragraph element for blocks of text.
- `<br>` — Line break (self-closing, no closing tag).
- `<hr>` — Horizontal rule / thematic break.
- `<strong>` — Bold / strong importance.
- `<em>` — Italic / emphasized text.
- `<mark>` — Highlighted / marked text.
- `<small>` — Smaller side text.
- `<sub>` and `<sup>` — Subscript and superscript.

Tip: Use Semantic Tags Over Style Tags

Prefer `<strong>` over `<b>` and `<em>` over `<i>`. Semantic tags convey **meaning** to browsers and assistive technologies, not just visual style.

Warning: Heading Hierarchy

Do not skip heading levels (e.g., jumping from h1 to h3). Screen readers and SEO rely on proper heading hierarchy for structure and navigation.

Key takeaway: Use the right text elements for the right purpose — headings for structure, paragraphs for content, and semantic tags for emphasis.

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Topic 4: Links and Images

Links and images are fundamental to making web pages interactive and visually rich. Links allow navigation between pages, while images bring content to life.

Key Concepts — Links

- `<a>` — The anchor element creates hyperlinks.
- `href` — Specifies the URL the link points to.
- `target="_blank"` — Opens link in a new tab.
- `<a href="#section">` — Links to an element with matching `id` on the same page.
- Absolute vs. relative URLs — `https://example.com` vs. `./about.html`.

Key Concepts — Images

- `<img>` — Embeds an image (self-closing tag).
- `src` — Path or URL to the image file.
- `alt` — Alternative text for accessibility and SEO.
- `width` / `height` — Sets display dimensions.
- Supported formats: JPG, PNG, GIF, SVG, WebP.

Tip: Always Include Alt Text

The `alt` attribute is essential for **accessibility**. Screen readers use it to describe images, and browsers display it when images fail to load.

Warning: Broken Images

If the `src` path is incorrect, the image will not display. Always verify file paths and use descriptive filenames.

Key takeaway: Links connect pages; images enrich them. Always use meaningful href values and descriptive alt text.

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Topic 5: Lists

Lists organize information into structured, readable formats. HTML provides three types of lists: ordered, unordered, and definition lists.

Key Concepts

- `<ul>` — Unordered list (bullet points).
- `<ol>` — Ordered list (numbered items).
- `<li>` — List item (used inside ul and ol).
- `<dl>` — Description / definition list.
- `<dt>` — Term in a description list.
- `<dd>` — Description of the term.
- Lists can be **nested** — a list inside a list item creates sub-lists.
- `type` attribute on `<ol>` — Changes numbering style (1, A, a, I, i).

Tip: Choose the Right List Type

Use **ordered lists** when sequence matters (steps in a recipe, instructions). Use **unordered lists** when items have no particular order (features, bullet points).

Warning: Nesting Lists Correctly

When nesting, the inner list must be placed **inside** an `<li>` element, not between list items. Incorrect nesting leads to invalid HTML.

Key takeaway: Lists structure information clearly. Master all three types and understand how to nest them properly.

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Topic 6: Tables

Tables display data in rows and columns. They are essential for presenting structured information like schedules, comparisons, and data sets.

Key Concepts

- `<table>` — Container for the entire table.
- `<thead>` — Groups header rows.
- `<tbody>` — Groups body/data rows.
- `<tr>` — Table row.
- `<th>` — Table header cell (bold, centered by default).
- `<td>` — Table data cell.
- `colspan` — Merges cells horizontally.
- `rowspan` — Merges cells vertically.
- `<caption>` — Adds a table title/caption.
- `<tfoot>` — Groups footer rows.

Tip: Use thead

Grouping rows with `<thead>`, `<tbody>`, and `<tfoot>` improves accessibility and makes it easier to style different sections of the table.

Warning: Tables for Data

Never use tables for page layout. Tables should only present **tabular data**. For layout, use CSS (Flexbox, Grid).

Key takeaway: Tables organize data into rows and cells. Use semantic grouping and avoid tables for layout purposes.

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Topic 7: Forms and Input

Forms are how users interact with websites — submitting data, making selections, and providing input. They are the backbone of login pages, registrations, surveys, and more.

Key Concepts

- `<form>` — Container for all form elements.
- `action` — URL where form data is sent on submit.
- `method` — HTTP method: `GET` or `POST`.
- `<input>` — The most versatile form element.
- Input types: `text`, `password`, `email`, `number`, `date`, `checkbox`, `radio`, `file`, `submit`, `reset`.
- `<label>` — Labels for inputs (improves accessibility).
- `<select>` — Dropdown selection menu.
- `<option>` — Individual option inside a select.
- `<textarea>` — Multi-line text input.
- `<fieldset>` and `<legend>` — Group related form elements.
- `placeholder`, `required`, `disabled` — Common input attributes.

Tip: Always Use Labels

Pair every input with a `<label>` using the `for` attribute. This makes forms accessible and allows users to click the label to focus the input.

Warning: GET vs. POST

Use `method="GET"` for searches and data retrieval. Use `method="POST"` for sensitive data, file uploads, and form submissions that modify server state. Never send passwords via GET.

Key takeaway: Forms collect user input. Use the right input types, always include labels, and choose the correct HTTP method.

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Topic 8: Semantic HTML

Semantic HTML uses elements that clearly describe their meaning to both browsers and developers. Instead of generic divs, semantic tags tell you **what** content is, not just how it looks.

Key Concepts

- `<header>` — Page or section header.
- `<nav>` — Navigation links.
- `<main>` — Primary content of the page.
- `<article>` — Self-contained content (blog post, news article).
- `<section>` — Thematic grouping of content.
- `<aside>` — Sidebar / tangential content.
- `<footer>` — Page or section footer.
- `<figure>` and `<figcaption>` — Self-contained media with caption.
- `<time>` — Machine-readable date/time.
- `<address>` — Contact information.

Tip: Replace Divs with Semantic Tags

Wherever possible, replace `<div>` with semantic elements. A `<div>` tells you nothing about its content; `<nav>` tells you it contains navigation.

Warning: Accessibility Matters

Screen readers and assistive technologies rely on semantic elements to navigate pages. Using non-semantic markup makes your site harder to access for users with disabilities.

Key takeaway: Semantic HTML improves accessibility, SEO, and code readability. Always choose meaningful elements over generic divs.

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Topic 9: HTML Media

Modern web pages are rich with media — audio, video, and embedded content from external sources. HTML5 introduced native elements for media without requiring plugins.

Key Concepts

- `<audio>` — Embeds audio content.
- `<video>` — Embeds video content.
- `<source>` — Specifies media files with different formats.
- `controls` — Displays playback controls for audio/video.
- `autoplay` — Starts playing automatically (use sparingly).
- `loop` — Repeats playback continuously.
- `muted` — Mutes the media by default.
- `<iframe>` — Embeds external content (YouTube, maps, etc.).
- Supported formats: MP3, WAV, OGG (audio); MP4, WebM, OGG (video).

Tip: Always Provide Controls

Never embed audio or video without the `controls` attribute. Users should always have the ability to play, pause, and adjust volume.

Warning: Autoplay Can Annoy

Avoid using `autoplay` — especially with audio. Most browsers now block autoplay with sound. It creates a poor user experience and can cause accessibility issues.

Key takeaway: HTML5 provides native media elements. Always include controls and multiple source formats for cross-browser compatibility.

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★ Topic 10: Final Project — Developer Profile Website

This final project brings together everything you have learned throughout the course. You will build a complete, multi-section **Developer Profile Website** using HTML.

Project Overview:
The Developer Profile Website showcases a developer's skills, projects, and contact information. It is a practical, real-world application of every HTML concept covered in this course.

Project Requirements

- Use proper HTML5 document structure (DOCTYPE, head, body).
- Include semantic elements: `<header>`, `<nav>`, `<main>`, `<section>`, `<footer>`.
- Add text formatting: headings, paragraphs, strong/em tags.
- Include links (navigation and external).
- Display images with descriptive alt text.
- Use lists for skills and project features.
- Create a table for the development timeline or tech stack.
- Build a contact form with various input types.
- Embed media (project demo video or audio intro).
- Apply semantic markup throughout for accessibility.

Tip: Build Incrementally

Start with the basic structure, then add sections one at a time. Test your page in the browser after each section to catch issues early.

Warning: Validate Your HTML

Before finishing, run your HTML through the **W3C Validator** (validator.w3.org). Valid HTML ensures cross-browser compatibility and better accessibility.

Sections to Include:

1. Header with name and navigation
2. About Me section with bio and photo
3. Skills section with lists
4. Projects section with descriptions and links
5. Tech Stack table
6. Contact form
7. Footer with social links

This project is your chance to demonstrate mastery of HTML. Take your time, be creative, and make it your own!